Name _____ Period _____ Date _____

2-3 Lesson Quiz: Factored Form of a Quadratic Function

short form Questions keep their Savvas quiz numbers. Show your work.

1. Factor the expression $3x^2 - 11x - 4$.

(_____ x + _____)(x − _____)

Work:

2. Solve the equation $x^2 + 7x = 30$.

Ⓐ $x = 15$ and $x = -2$

Ⓑ $x = -6$ and $x = 5$

Ⓒ $x = -10$ and $x = 3$

Ⓓ $x = 10$ and $x = -3$

Show the factored equation and the Zero Product Property step:

5. A long-jumper lifts off 3 m after starting his run, and lands 6 m from where he lifted off. When he is 8 m from the start line, he is 5 cm above the ground. Write the equation of a parabola that models his path through the air, where x is his horizontal distance from the start line in m and y is his height, in cm.

Ⓐ $y = -x^2 + 12x - 27$

Ⓑ $y = -3x^2 + 4x - 9$

Ⓒ $y = 3x^2 - 4x + 9$

Ⓓ $y = x^2 - 12x + 27$

Zeros: $x =$ _____ and $x =$ _____

Point: (_____, _____)

Work:

Scoring: Question 2 and Question 5 are 10 points each. Question 1 is a bonus of up to 5 points. Total out of 20.

2-3 Lesson Quiz, short form: answer key

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1. **3, 1, 4:** $(3x+1)(x-4)$**.** $ac = 3(-4) = -12$; the factors 1 and -12 have sum -11. $3x^2 + x - 12x - 4 = x(3x+1) - 4(3x+1) = (3x+1)(x-4)$. Check: $3x^2 - 12x + x - 4 = 3x^2 - 11x - 4$. Savvas quiz item 1, DOK 2, Skills Review A24, A69. Taught by Example 1, Part B (cut); the packet rehearses the grouping step in Example 3, Part B. **Bonus 5:** all three blanks correct; 3 for the correct binomials with a sign slip; 0 otherwise.

2. **C,** $x = -10$ **and** $x = 3$**.** $x^2 + 7x - 30 = 0$, $(x+10)(x-3) = 0$, so $x = -10$ or $x = 3$. Distractor B solves $x^2 + x = 30$; D has both signs flipped, the error of reading the constants in $(x+10)(x-3)$ as the zeros. Savvas quiz item 2, DOK 2, A24. Example 3, Part A, and Try It 3a. **10 points:** 6 for C, 4 for the factored equation and the Zero Product Property step.

5. **A,** $y = -x^2 + 12x - 27$**.** Lift-off at $x = 3$; he lands 6 m *after* lift-off, at $x = 3+6 = 9$. So $y = a(x-3)(x-9)$. The point $(8, 5)$: $5 = a(5)(-1)$, $a = -1$. $y = -(x-3)(x-9) = -x^2 + 12x - 27$. Trap: using 6 as the second zero. D has $a = +1$ (opens up, below ground between the zeros). Savvas quiz item 5, DOK 2, F50, F58. Example 6 and the packet Practice (Practice & Problem Solving 45). **10 points:** 4 for the zeros 3 and 9, 2 for the point $(8, 5)$, 4 for A.

Items left off this form (available for a retake or Form B):

- Savvas item 3, the projectile $h(t) = -16t^2 + 32t + 128$: answer 4 seconds ($-16(t-4)(t+2) = 0$, and time cannot be negative), DOK 2, A29; it is taught by Example 4 (cut) and its function is the same as Try It 4's, so it cannot be a fresh item for students who did Try It 4.
- Savvas item 4, intervals where $f(x) = x^2 + 12x + 27$ is positive: answer -9, -3, that is $(-\infty, -9)$ and $(-3, \infty)$, DOK 2, A70; taught by Example 5 (cut).

IXL replacement: SmartScore 70 or above on UB5, CJC and HPB (the starred skills; UB5 feeds the item 1 bonus) replaces this quiz with 20/20.

Verify every key against the Savvas Realize quiz PDF (aga_24_a2_na_0203_lq.pdf) on quiz day; this form is built from the TE transcription (a2_2-3_TE.tex, printed page 82A).